S.Y. B. Tech. Academic Year 2026_27 Semester: III

CSF20030Data Structures –I Laboratory

LABORATORY WRITE UP

Experiment Number: 09

TITLE: Job Queue Simulation Using Linear Queue

PROBLEM STATEMENT:

Queues are frequently used in computer programming, and a typical example is the creation of a job queue by an operating system. If the operating system does not use priorities, then the jobs are processed in the order they enter the system. Write a C program for simulating a job queue. Write functions to add a job and delete a job.

OBJECTIVE:

1. Understand the concept of Queue and FIFO (First In, First Out).
2. Implement a linear queue using an array.
3. Implement functions to add and delete jobs.
4. Simulate processing of jobs in the order they enter the system.
5. Analyze the time and space complexity of queue operations.

IMPLEMENTATION:

- **PLATFORM:**
  - 64-bit Open source Linux or its derivatives.
  - Open Source C Programming tool like gcc/Eclipse Editor.

- **TEST CONDITIONS:-**

Test Case	Input	Expected Output
TC01	Add one job	Job added successfully
TC02	Add multiple jobs	Jobs stored in FIFO order
TC03	Delete a job	First entered job is deleted/processed
TC04	Delete from empty queue	Queue Underflow message
TC05	Add job when queue is full	Queue Overflow message
TC06	Display pending jobs	All jobs displayed from front to rear

- **PSEUDO CODE:** ***//To be Written by Students***

  To be written by students: Initialize front and rear, accept menu choice, add jobs at the rear, delete/process jobs from the front, display pending jobs, and handle overflow/underflow conditions.
- **TIME COMPLEXITY:** ***//To be Written by Students***

  Write time complexity of above operations
- **CONCLUSION:**

  The job queue simulation demonstrates the FIFO principle of a linear queue. Jobs are added at the rear and processed/deleted from the front in the same order in which they enter the system.

FAQs ***//To be Written by Students***

1. What is a Queue?
2. What is FIFO?
3. Why is a queue suitable for an operating-system job queue?
4. What is Queue Overflow and Queue Underflow?
5. What are the applications of Queue?
6. What is a Circular Queue?
7. Why is a Circular Queue preferred over a Linear Queue?
8. What principle does a Circular Queue follow?
9. What are the basic operations performed on a Circular Queue?
10. What is an overflow & Underflow condition in a Circular Queue?

Practice Assignments – Circular Queue

1. Implement the job queue using a circular queue.
2. Add a function to display the number of pending jobs.
3. Implement a function to view the front job without deleting it.
4. Store job names along with job IDs.
5. Implement a priority queue for jobs with different priorities.